Ethan Miao

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Education

University of Washington — Seattle, WA
M.S. in Electrical and Computer Engineering — *Sep 2026 – Expected Jun 2028*

Huazhong Agricultural University — Wuhan, China
B.S. in Computer Science and Technology; GPA: 3.94/4.00 — *Sep 2022 – Jun 2026*

Professional Experience

ZWSOFT — Wuhan, China
Software Development Engineer Intern — *May 2026 – Aug 2026*

- Developed a **C++ AI Assistant plugin** for ZWCAD and designed an asynchronous IPC layer between CAD and an independent AI service for CAD drawing processing and knowledge retrieval; added structured logging and end-to-end Request ID tracing for cross-process observability, troubleshooting, and debugging.
- Built a **FastAPI** gateway for a distributed backend spanning retrieval, inference, knowledge-base, and internal services; implemented **REST/gRPC** routing and service-to-service communication, using PostgreSQL for persistent task state and Redis for caching, reducing retrieval P95 latency by **54%** from 460 ms to 210 ms.
- Engineered a high-concurrency pipeline using asynchronous I/O, **RabbitMQ**, and worker pools to decouple online requests from long-running parsing and indexing jobs; added connection pooling, rate limiting, and timeouts, sustaining 500+ concurrent requests and improving throughput by **27%**.
- Containerized backend services with **Docker** and deployed them to an internal **Kubernetes** cluster with load balancing, health checks, automatic restarts, and horizontal scaling; improved operational excellence with Prometheus and Grafana, reducing mean recovery time from **10 to 6 minutes**.

Jinghua Technology Co., Ltd. — Wuhan, China
Software Development Engineer Intern — *Jul 2025 – Nov 2025*

- Built a responsive web frontend with **React and TypeScript** and implemented role-based access control with **JWT/OAuth2** to manage permissions across different user roles.
- Built RESTful backend services with FastAPI, containerized with Docker, and deployed to **AWS EC2** behind **Nginx** reverse proxies and load balancers; supported **1K+** concurrent requests under load testing.
- Hardened REST APIs with **idempotent writes**, rate limiting, and structured logging; created API contracts and technical documentation with **Swagger** to communicate effectively, reducing duplicate-write errors by **20%**.
- Built a **GitHub Actions** CI/CD pipeline across the software development lifecycle, with automated linting, unit and integration tests, Docker image builds, and deployments, reducing manual deployment effort.

Projects

Distributed Cloud Storage Platform

- Architected a multi-tenant cloud storage platform using Factory and Strategy design patterns, with **7** Spring Boot microservices and **40+** REST APIs, deployed on AWS EC2 behind ALB, sustaining **300+** RPS.
- Built a resumable large-file upload pipeline with S3 multipart uploads, presigned URLs, Redis distributed locks, checksums, and deduplication, supporting files up to **20 GB** without retransmitting completed chunks.
- Designed a near-real-time file search pipeline that streamed metadata changes from PostgreSQL to OpenSearch via Debezium, indexing **1M+** records and reducing search latency by **65%** over indexed SQL queries.
- Built a **RAG-powered** AI agent for natural-language file search, document summarization, cross-document Q&A, and file operations; exposed search, metadata lookup, and document retrieval as standardized **MCP** tools, orchestrated with LangGraph, and used Kafka with retries and DLQs for asynchronous parsing and indexing.

Skills

Programming Languages: C, C++, C#, Python, Java, Rust, JavaScript, CSS, TypeScript, SQL
Backend & Distributed Systems: FastAPI, REST APIs, gRPC, PostgreSQL, MySQL, Redis (NoSQL), RabbitMQ
Cloud Services & DevOps: AWS, Docker, Kubernetes, Nginx, GitHub Actions, Prometheus, Grafana, Linux, Git
AI & Machine Learning: AI Agents, RAG, MCP, Vector Search, LangChain, LangGraph, PyTorch, Fine-tuning
Professional Skills: Written and Verbal Communication, Collaboration, Analytical Problem Solving

Publications

Wang S, Zhao S, Miao S, Kwashie S, Bewong M, Hu J, Nofong VM, Feng Z. FastER: On-demand Entity Resolution in Property Graphs. International Semantic Web Conference 2025 Oct 29 (pp. 177-195). Cham: Springer Nature Switzerland.